

GENERAL OZochiawaeze

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EDUCATION	Purdue University <i>Ph.D. Candidate in Mathematics</i> 2021 – 2027 (expected) • Research Interests: Inverse Problems & Partial Differential Equations • Advisors: Peijun Li & Isaac Harris
	New Jersey Institute of Technology <i>M.S. in Applied Mathematics</i> May 2021 • Advisor: Christina Frederick
	Rutgers University <i>B.A. in Mathematics</i> May 2018
PUBLICATIONS & PREPRINTS	1. R. Ayala, I. Harris, G. Ozochiawaeze. Factorization method for the biharmonic scattering problem for an absorbing penetrable obstacle. <i>Submitted</i>, 2025. arXiv:2511.05711 2. I. Harris, P. Li, G. Ozochiawaeze. Sampling methods for the inverse cavity scattering problem of biharmonic waves. <i>Under Review, Inverse Problems</i>, 2025. arXiv:2509.02773
WORK EXPERIENCE	MIT Lincoln Laboratory Lexington, MA May 2023 – Oct 2023 • <i>Summer Research Intern, Group 36 – Integrated Missile Defense Technology</i> • Applied compressive sensing to enhance radar imaging for missile target detection, improving reconstruction accuracy. • Presented compelling research findings on compressed radar sensing, demonstrating potential applications in enhanced defense technologies.
	MIT Lincoln Laboratory Lexington, MA May 2022 – Aug 2022 • <i>Summer Research Intern, Group 37 – Advanced Undersea Systems & Technology</i> • Developed algorithms to analyze coherence loss in sonar signals, improving underwater detection and tracking. • Applied sonar array interferometry to extract spatial information from acoustic wave-fronts.
	New Jersey Institute of Technology Newark, NJ Jul 2020 – May 2021 • <i>Acoustic Simulation Researcher</i> • Devised oceanographic models to simulate acoustic wave propagation in underwater environments in COMSOL. • Critically applied finite element methods and domain decomposition modeling to simulate high-frequency acoustic wave behavior in complex ocean conditions.
TEACHING	• Lecturer, Purdue University — MA 16020: <i>Applied Calculus II</i>, Fall 2025 • Recitation Instructor, Purdue University — MA 261: <i>Multivariable Calculus</i>, Spring 2024 • Teaching Assistant, Purdue University — MA 511: <i>Linear Algebra and Applications</i>, Spring 2023 • Recitation Instructor, Purdue University — MA 261: <i>Multivariable Calculus</i>, Fall 2022 • Recitation Instructor, Purdue University — MA 162: <i>Calculus II</i>, Spring 2022 • Recitation Instructor, Purdue University — MA 161: <i>Calculus I</i>, Fall 2021 • Instructor, Accel Learning — Mentored students in preparation for AMC and AIME math competitions, Spring–Fall 2020 • Math Instructor, Mathnasium, K-12 Teaching & Tutoring, Fall 2018–Spring 2019

AWARDS AND HONORS	• SIAM (Society of Industrial & Applied Mathematics) Student Travel Award	2025
	• NSF Computational Mathematics Program Award, DMS-2208256,	2023-2024
	• GEM Fellowship	2022
	• Graduate C/Startup Research Grant	2020
	• Tau Sigma Honors Society	2016

SKILLS	Computing & Software: Python, C++, MATLAB, COMSOL, FEniCS
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SEMINAR & CONFERENCE TALKS	A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media (invited speaker)	2025
	SIAM Conference on Analysis of Partial Differential Equations (PD25), Pittsburgh	
	Localizing Cavities in Biharmonic Scattering with Limited Data (contributed paper)	2025
	SIAM Conference on Analysis of Partial Differential Equations (PD25), Pittsburgh	
	Localizing Cavities in Biharmonic Scattering with Limited Data	2025
	CCAM Lunch Seminar, Purdue University	
	Extending Qualitative Reconstruction to Biharmonic Scattering with Limited Data	2025
	Summer School: Theory and Applications of Elliptic PDE, UC Irvine	
	Extending Qualitative Reconstruction in Biharmonic Scattering with Limited Data	2025
	Graduate Student Analysis Seminar, Purdue University	
	On Sampling Methods for Recovering a Clamped Cavity in a Thin Plate	2025
	CCAM Lunch Seminar, Purdue University	
	A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate	2024
	Purdue Graduate Research Day, Purdue University	
	A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate	2024
	SIAM Student Chapter Conference, Purdue University	
	Comparing and Integrating the Probe Method and Method of Singular Sources	2023
	Graduate Student Analysis Seminar, Purdue University	
	The Factorization Method in Inverse Scattering	2023
	Graduate Student Analysis Seminar, Purdue University	
	Compressive Radar Imaging in Missile Defense	2023
	Integrated Missile Defense Technology Group, MIT Lincoln Laboratory	
	Quantifying Real Operational Performances of SONAR Arrays in Random Ocean environments	2022
	Advanced Undersea Systems & Technology Group, MIT Lincoln Laboratory	
	Finite Element Modeling of Underwater Acoustic Environments	2022
	Purdue Math Student Colloquium, Purdue University	

CONFERENCES ATTENDED	SIAM Conference on Analysis of Partial Differential Equations (PD25), Pittsburgh	2025
	Pittsburgh, PA	
	Summer School: Theory and Applications of Elliptic PDE	2025
	UC Irvine	
	Mathematics and Statistics in Industry Workshop and Panel Discussion	2025
	Purdue University	
	Conference on Mathematics of Wave Phenomena	2025
	Karlsruhe Institute of Technology (KIT)	
INFORMAL TALKS	Paris-Saclay Conference in Analysis and PDE	2024
	Laboratoire de Mathématiques d'Orsay	
	SIAM Student Chapter Conference	2024
	Purdue University	
	Workshop on Scientific Computing	2022
	Purdue University	
	New Ideas in Computational Inverse Problems	2022
	BIRS, Banff International Research Station	
	The Dawn of the Diophantine Era: Algebra at the Crossroads of Empires	2025
	<i>Math History Seminar, Purdue University</i>	
	Fixed Points in Action: from Brouwer to Lefschetz & Beyond	2025
	<i>Topologies, Purdue University</i>	
	The Life, Mathematics, and Political Advocacy of Sofya Kovalevskaya	2024
	<i>Math History Seminar, Purdue University</i>	
	The K-theoretic Atiyah-Singer Index Theorem	2024
	<i>Student Operator Algebras Seminar, Purdue University</i>	
SERVICE	Introducing Pseudodifferential Operators	2024
	<i>Purdue Math Student Colloquium</i>	
	Bourbaki and the Foundations of Modern Math	2023
	<i>Math History Seminar, Purdue University</i>	
	Wasan: Mathematical Treasures in Early Modern Japan	2022
	<i>Math History Seminar, Purdue University</i>	
	Purdue Math Department Graduate Representative	2025-2026
	<i>Purdue University</i>	
	Mentor, Directed Reading Program in Computational Topology	2025
	<i>Purdue University — mentee: Aaron (Sang Hyun) Kim</i>	
	Founder and Organizer, Math History Seminar	2022
	<i>Purdue University</i>	
	Rutgers Undergraduate Mathematics Association Treasurer & Web Manager	2017-2018
	<i>Rutgers University</i>	